Exploring the shapes of graphs using technology

1 The linear equation $y = 3x - 7$ will form a __________ graph. (Tick **1** correct answer)

- ☐ curved
- ☐ decreasing
- ☑ straight line

2 Which of these coordinates will be on the linear graph $y = 3x - 7$? (Tick **3** correct answers)

- ☑ (5, 8)
- ☐ (8,5)
- ☐ (-4, -1)
- ☑ (1, -4)
- ☑ (2, -1)

3 How many points can be plotted on the graph of the linear equation $y = 3x - 7$? (Tick **1** correct answer)

- ☐ All integers.
- ☐ Positive integer values only.
- ☐ Negative integer values only.
- ☐ Millions but, there is a limit.
- ☑ The number of points you can plot on this graph goes on infinitely.

4 If you were populating this table of values for the linear equation $y = 8 - 5x$, what is highest value of y you will get? (Tick **1** correct answer)

x	-3	-2	-1	0	1	2	3
y							

- [ ] 3
- [ ] 5
- [ ] 8
- [x] 23
- [ ] 28

5 If you were populating this table of values for the linear equation $y = 8 - 5x$, what is lowest value of y you will get? (Tick **1** correct answer)

x	-3	-2	-1	0	1	2	3
y							

- [ ] 23
- [ ] -3
- [ ] -5
- [x] -7
- [ ] -23

6 Which of these equation will produce a non-linear graph? (Tick **1** correct answer)

- [ ] $y = 3x + 8$
- [ ] $y = 3x - 8$
- [ ] $y = 8 - 3x$
- [ ] $3x + y = 8$
- [x] $y = x^3 + 8$